

Project-1 Mental Health in Tech Survey using SQL

Project Title: Mental Health Analytics in the Technology Industry Using SQL, Excel, and Power BI

1. Project Overview

Mental health has become one of the most important workplace concerns in the technology industry. High-pressure environments, long working hours, remote work culture, and job-related stress can significantly impact employee well-being and productivity.

This capstone project focuses on analyzing the Mental Health in Tech Survey dataset to uncover patterns, trends, and insights related to mental health conditions, workplace support systems, and employee behavior in technology companies.

Using tools such as:

- SQL
- Excel
- Power BI

students will perform:

- Data cleaning
- Exploratory Data Analysis (EDA)
- KPI analysis
- Dashboard development
- Business insight generation
- Data storytelling

Objective:

This project is designed to simulate a real-world healthcare and HR analytics use case where organizations want to understand:

- Employee mental health trends
- Impact of workplace policies
- Treatment-seeking behavior
- Organizational support effectiveness

The objective of this project is to analyze survey responses and generate actionable insights that organizations can use to improve employee wellness initiatives.

3. Project Learning

The main learning of this capstone project are:

Analytical:

- Analyze employee mental health trends
- Understand treatment patterns
- Identify workplace risk factors
- Evaluate company support systems
- Compare mental health behavior across demographics

Technical:

- Perform SQL-based analysis
- Create interactive Power BI dashboards
- Use Excel for exploratory analysis
- Build KPI reports
- Generate business recommendations

4. Dataset Description

Dataset Name:

Mental Health in Tech Survey

Dataset Type:

Survey Dataset

Domain:

Healthcare Analytics / HR Analytics

Data Format:

CSV File

Data Source:

[Public mental health workplace survey data](#)

5. Dataset Features / Columns Description

Column Name	Description
Timestamp	Survey submission timestamp
Age	Respondent age
Gender	Gender of respondent
Country	Country of respondent
State	State information
self_employed	Whether self-employed
family_history	Family history of mental illness
treatment	Whether respondent sought treatment
work_interfere	Mental health interference with work
no_employees	Company size
remote_work	Remote work status

Column Name	Description
tech_company	Whether company is tech-related
benefits	Availability of mental health benefits
care_options	Availability of care options
wellness_program	Wellness program availability
seek_help	Resources for seeking help
anonymity	Whether anonymity is protected
Leave	Ease of taking leave for mental health
mental_health_consequence	Fear of negative consequences
phys_health_consequence	Physical health discussion consequences
coworkers	Comfort discussing with coworkers
supervisor	Comfort discussing with supervisor
mental_health_interview	Discuss mental health in interview
phys_health_interview	Discuss physical health in interview
mental_vs_physical	Importance of mental vs physical health
obs_consequence	Observed negative consequences

6. Scope

The scope of this project covers:

- Data cleaning and preprocessing
- SQL querying
- Data visualization
- Dashboard creation

- Trend analysis
- HR analytics reporting
- Business recommendation generation

7. Tech Stack

Tool	Purpose
SQL	Data querying and analysis
Excel	Initial cleaning and pivot analysis
Power BI	Dashboard and visualization
Python or R (Optional)	Advanced analytics
Tableau (Optional)	Alternative dashboarding

8. Data Cleaning Tasks

Perform the following preprocessing tasks:

Data Quality Checks

- Remove duplicate records
- Handle null values
- Standardize gender values
- Remove unrealistic age values
- Standardize categorical values

Example Cleaning Rules

- Age should be between 18 and 65
- Merge similar gender categories
- Replace blank responses with NULL

- Convert Yes/No columns into standardized format

9. Key Performance Indicators (KPIs)

Employee Wellness KPIs

- Total Respondents
- Average Employee Age
- Treatment Percentage
- Remote Work Percentage
- Employees with Wellness Programs
- Employees with Mental Health Benefits
- Employees Comfortable Discussing Mental Health

Organizational KPIs

- Support Program Coverage
- Leave Accessibility Score
- Mental Health Risk Index
- Employee Support Index

10. SQL Analysis Questions

Basic Analysis

1. Total number of respondents
2. Average age of employees
3. Gender distribution analysis
4. Country-wise respondent count
5. Percentage of remote workers

Intermediate Analysis

6. Treatment analysis by country
7. Company size distribution
8. Mental health benefits availability
9. Wellness program analysis
10. Family history vs treatment analysis

Advanced Analysis

11. Employees afraid to discuss mental health
12. Mental health interference analysis
13. Treatment rate by gender
14. Leave policy effectiveness
15. Supervisor support analysis

Data Science Oriented Analysis

16. High-risk employee segmentation
17. Correlation between benefits and treatment
18. Predictive dataset preparation
19. Country-wise risk scoring
20. Employee support clustering logic

11. Power BI Dashboard Requirements

Dashboard 1: Executive Overview

Visuals:

- KPI cards
- Country map
- Treatment trend

- Gender analysis
 - Company size analysis
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Dashboard 2: Employee Wellness Insights

Visuals:

- Work interference heatmap
 - Treatment analysis
 - Wellness support availability
 - Leave accessibility analysis
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Dashboard 3: Organizational Risk Dashboard

Visuals:

- Risk score analysis
- Support index
- Remote work comparison
- Mental health consequence analysis

12. Business Insights Expected

You have to submit a report that includes your findings. Please include the points below in your report.

- Countries with higher treatment rates
- Effectiveness of wellness programs
- Workplace support gaps
- Employee behavior trends
- Relationship between company support and treatment

13. Recommendations

Students should provide recommendations based on their analytics and technical analysis of the survey data provided. Please include the points below in your recommendation.

- Improve employee wellness programs
- Increase awareness initiatives
- Enhance anonymous support systems
- Train managers on mental health awareness
- Improve leave accessibility

Additional Task (Optional)

Machine Learning Ideas

- Predict whether an employee seeks treatment
- Employee risk classification
- Sentiment/risk scoring

Statistical Analysis

- Correlation analysis
 - Hypothesis testing
 - Regression analysis
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14. Deliverables

You should submit:

SQL Deliverables

- SQL scripts
- Analytical queries
- Views/stored procedures (optional)

Excel Deliverables

- Pivot tables
- Charts
- Data cleaning workbook

Power BI Deliverables

- Interactive dashboard
- KPI report
- Drill-down analysis

Documentation

- Final project report
- Business recommendations
- Presentation slides

15. Learning Outcomes

After completing this project, students will be able to:

- Perform real-world healthcare analytics
- Write analytical SQL queries
- Build interactive dashboards
- Generate business insights
- Work on end-to-end analytics projects
- Apply data storytelling techniques

Suggested Project Timeline

Phase	Duration
Data Understanding	1 Day
Data Cleaning	1 Day
SQL Analysis	2 Days
Dashboard Development	2 Days
Insight Generation	1 Day
Final Presentation	1 Day

Resume/Portfolio Value

Suitable for:

- Data Analyst roles
- Business Analyst roles
- Healthcare Analytics roles
- BI Developer positions
- Junior Data Scientist positions